# CHARACTERIZING CALCYANIN, AN ORPHAN PROTEIN POTENTIALLY INVOLVED IN BIOMINERALIZATION

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Cyanobacteria are photosynthetic microorganisms that fix atmospheric $CO_2$ and bicarbonate to produce sugars, generating calcium carbonate ($CaCO_3$) as a by-product, mostly extracellularly, but intracellularly in some species. This latter trait is phylogenetically widespread, and some species, such as *Gloeomargarita lithophora*, preferentially precipitate heavy alkaline-earth elements such as strontium and radium[1], opening some perspectives for the bioremediation of radioactive isotopes. The biomineralization byproducts are enclosed within an envelope of still-undetermined nature (lipidic and/or protein). Comparative genomics identified a single gene, *ccyA*, present in intracellularly biomineralizing strains and absent from non-biomineralizing ones[2]. This gene has also been acquired by horizontal gene transfer in methanotrophic gammaproteobacteria[3]. *ccyA* encodes an uncharacterized, orphan protein named calcyanin, showing no sequence similarity to any protein of known function. Calcyanin contains two distinct domains: a variable N-terminal domain and a conserved C-terminal domain, made of three consecutive, long glycine-zippers (« GlyZip3 » domain)[2].

In this study, we aim to understand calcyanin's function(s) and its role in the biomineralization process using a multidisciplinary approach, combining bioinformatics with experimental characterization.

The *in silico* investigations we are conducting involve searching for the presence of the calcyanin's various domains in other proteins with diverse structures, with the aim of guiding their experimental characterization. A recent study, combining searches for distant sequence relationships with structural predictions provided by AlphaFold2, led to the description of a new family of domains (CoBaHMA)[4], found within the N-terminal region of some calcyanins and various proteins linked to membranes or involved in lipid metabolism. Here, we focus on the core C-terminal region of calcyanins, which consists of long glycine zippers forming helix hairpins. These helices are closely associated at the level of glycine repeated every four residues, while a conserved glycine-proline dipeptide forms the hairpin. We searched for distant homologs among proteins other than calcyanins, by comparing sequence profiles that are built from close homologs of each GlyZip's module, against a database of profiles representative of homologous families. We used HHblits iteratively and transitively to capture GlyZip's sequences. We further annotated the GlyZip superfamily identified in this way with data coming from protein sequence classification ressources (InterPro) as well as from structure databases (AlphaFold, PDB). For experimental structural characterization of calcyanin, we express the *ccyA* gene of *Gloeomargarita lithophora* and *Methylococcus geothermalis*in *E. coli* C41/C43. Both calcyanins, with production improved by calcium supplementation, were purified from soluble and membrane fractions by affinity and gel-filtration chromatography, and confirmed by Western Blot (anti-HIS, anti-STREPT). Negative-stain EM and mass spectrometry are ongoing. Large-volume cultures will next supply protein to purify further constructs for negative-stain EM and, subsequently, cryo-EM. Separately, intracellular carbonate ultrastructure is studied by cryo-FIB milling and cryo-ET, in collaboration with the Cryo-EM facility at IBS, Grenoble.

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